

Check Your Understanding of Topic 1.4

OP Questioning and Predicting **PC** Planning and Conducting **PA** Processing and Analyzing **E** Evaluating
AI Applying and Innovating **C** Communicating

Understanding Key Ideas

1. Use a flowchart with diagrams to summarize gene cloning. **PA C**
2. What is the benefit of producing insulin from transgenic plants rather than transgenic bacteria? **PA**
3. A company has developed a transgenic carrot that secretes toxins that kill damaging insects and worms. **AI C**
 - a) What are some of the risks and benefits that you think the Canadian government should consider when deciding whether to approve this plant for agricultural use?
 - b) If approved, what advantages will this transgenic carrot offer to farmers? What are some of the potential drawbacks to farmers?
 - c) Do you think that foods produced with genetically modified ingredients should be labelled so that consumers can make informed choices? List your arguments.
4. Explain the steps involved in in vitro fertilization. **PA C**
5. Use a graphic organizer of your choice to help explain how gene therapy works. **PA C**
6. Copy the following table into your notebook and complete it. **PA C**

Applications of Biotechnology

Application	Benefits	Risks/Concerns
Cloning		
Genetically modified organisms		
Gene therapy		

7. How can transgenic organisms help to achieve social, economic, or environmental goals? Give one example of a transgenic organism designed to meet one of these goals. **E C**

Connecting Ideas

8. Many people have life-threatening allergies to certain foods, such as nuts. Researchers are developing transgenic peanuts that will not produce allergic reactions. Do you think this is a good use of research money? Explain. **E C**

Making New Connections

9. Study the diagram below. What does it show? Explain your reasoning. **PA C**

